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import os
import time
from openai import OpenAI
import google.generativeai as genai
from transformers import pipeline

OPENAI_API_KEY = "sk-proj-4YP_Jjc9Av_oo0jIXB0MvWXZf1MBqT9uf3vH3vrmG1L2hLfELNb1NbWJRcJ7MK9u-WNmEo0sGpT3B1bkFJRWJtNyKWdQSYD9tc9FF0"
GOOGLE_API_KEY = "AIzaSyCXeD1XY-Ulgwte9f3Q76KApDpBNgXEuAY"

openai_client = OpenAI(api_key=OPENAI_API_KEY)

hf_generator = pipeline("text-generation", model="gpt2")

def query_openai(prompt):
    try:
        response = openai_client.chat.completions.create(
            model="gpt-4o-mini",
            messages=[{"role": "user", "content": prompt}],
            temperature=0.7
        )
        return response.choices[0].message.content.strip()
    except Exception as e:
        return f"OpenAI Error: {e}"

def query_gemini(prompt):
    try:
        model = genai.GenerativeModel("gemini-1.5-flash", api_key=GOOGLE_API_KEY)
        response = model.generate_content(prompt)
        return response.text.strip()
    except Exception as e:
        return f"Gemini Error: {e}"

def query_huggingface(prompt):
    try:
        result = hf_generator(prompt, max_length=60, num_return_sequences=1)
        return result[0]["generated_text"].strip()
    except Exception as e:
        return f"HuggingFace Error: {e}"

prompt = "Write a haiku about AI and nature."

print("\n=====")
print("PROMPT")
print("=====")
print(prompt)

responses = {}

models = {
    "OpenAI (gpt-4o-mini)": query_openai,
    "Gemini (1.5-flash)": query_gemini,
    "HuggingFace (GPT-2)": query_huggingface
}

for name, func in models.items():
    start = time.time()
    output = func(prompt)
    end = time.time()

    responses[name] = {
        "text": output,
        "time": round(end - start, 2)
    }

```

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print("\n\n=====")
print("MODEL RESPONSES")
print("=====\\n")

for model, data in responses.items():
    print(f"### {model}")
    print(f"Response Time: {data['time']} seconds\\n")
    print(data["text"])
    print("\\n-----\\n")

def evaluate_models(prompt, responses):
    evaluation_prompt = f"""
You are an unbiased evaluator.

Prompt:
{prompt}

Responses:
"""

    for model, data in responses.items():
        evaluation_prompt += f"\\nModel: {model}\\n{data['text']}\\n"

    evaluation_prompt += """
Rank the models from BEST to WORST based on:
- Creativity
- Relevance
- Fluency
- Poetic quality

Return only the model names in order.
"""

    try:
        response = openai_client.chat.completions.create(
            model="gpt-4o-mini",
            messages=[{"role": "user", "content": evaluation_prompt}],
            temperature=0
        )
        return response.choices[0].message.content.strip()
    except Exception as e:
        return f"Evaluation Error: {e}"

print("\\n\n=====")
print("FINAL RANKING")
print("=====\\n")

ranking = evaluate_models(prompt, responses)
print("🏆 Best → Worst:\\n")
print(ranking)

```

Loading weights: 100% 148/148 [00:00<00:00, 191.52it/s, Materializing param=transformer.wte.weight]

GPT2LMHeadModel LOAD REPORT from: gpt2

Key	Status	
h.{0...11}.attn.bias	UNEXPECTED	

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.

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PROMPT

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Write a haiku about AI and nature.
 Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.
 Both `max\_new\_tokens` (=256) and `max\_length` (=60) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to

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MODEL RESPONSES

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OpenAI (gpt-4o-mini)
 Response Time: 2.81 seconds

OpenAI Error: Error code: 429 - {'error': {'message': 'You exceeded your current quota, please check your plan and billing detai

Gemini (1.5-flash)
 Response Time: 0.0 seconds

Gemini Error: module 'google.generativeai' has no attribute 'GenerativeModel'

Start coding or [generate](#) with AI.

HuggingFace (GPT-2)
 Response Time: 3.55 seconds

Write a haiku about AI and nature.

"Yes, I know. I have been around in the video game industry for a long time, so I thought it would be interesting to share this

The video will run through the whole game.

"This is a game that is not for the faint-hearted. It is not for the faint-hearted to find a problem that can't be solved by the

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FINAL RANKING

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🏆 Best → Worst:

Evaluation Error: Error code: 429 - {'error': {'message': 'You exceeded your current quota, please check your plan and billing d